Title
Triage for critically ill patients under the limited medical resource capacity during the COVID-19 and other infectious disease pandemics: a scoping review.

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Abstract
Introduction
COVID-19 spread worldwide rapidly since December 12, 2019, affecting more than 180 countries. The rapidly increased number of patients sometimes exceed medical resource capacity. Under the scarce resource situation, clinicians may need to allocate medical resources to selected populations. However, there are no established triage criteria. Therefore, we conducted a scoping review to map the reported triage available. We formulated the following primary review question: What kind of criteria and algorithms are available for clinicians to triage patients in the resource scarcity setting during the disease pandemic like COVID-19?

Methods and analysis
For conducting the scoping review protocol, we will use a scoping review methodology established in the Joanna Briggs Institute manual. We also will use the Preferred Reporting Items for Systematic Reviews and Meta-Analyses Extension for Scoping Reviews to assess the results. The concept of our review will be the triage criteria, systems, or algorithms used for the allocation of medical resources to selected populations among patients with the pandemic disease. The context of our review will be the resource scarcity setting during disease pandemic like COVID-19. We will conduct the systematic search on MEDLINE, Embase, and Cochrane controlled register of trials (CENTRAL) and include studies that incorporate critically ill patients due to the pandemic disease itself. We will summarize the studies based on the characteristics and efficiency (if applicable) of triages.

Ethics and dissemination
Since this is a review of the literature, ethics approval is not mandated. We will publish the results from this review in peer-reviewed journals.

Introduction
COVID-19 spread worldwide rapidly since December 12, 2019, affecting more than 180 countries until April 2020. At the time of April 16, 2020, more than 1,480,000 cases and 130,000 deaths were reported.(1) COVID-19 induces acute respiratory distress syndrome and respiratory failure in severe cases. To save the critical patients, intensive care using advanced medical resources such as ventilator and extracorporeal membrane oxygenation (ECMO) is

necessary. However, the rapidly increased number of patients sometimes exceed medical resource capacity.(2,3) (Remuzzi 2020, Emanuel 2020)

Under the scarce resource situation, clinicians may need to allocate medical resources to selected populations. However, there are no established triage criteria as possible as we know. Therefore, we conducted the scoping review to map the reported triage available to support clinicians' decision making. We formulated the following primary review question:
1. What kind of criteria and algorithms are available for clinicians to triage patients in the resource scarcity setting during the disease pandemic like COVID-19?
We posed the following sub-questions:
2. Where and when was the triage required and done so far?
3. What were the benefits and harms of the triage?

Methods
We will conduct his scoping review according to the Preferred Reporting Items for Systematic Review and Meta-analysis extension for Scoping Reviews (PRISMA-ScR) and Joanna Briggs Institute Reviewer's manual.(4,5)

Eligibility criteria
Types of participants
We will include studies that incorporate critically ill patients due to the pandemic disease itself. We excluded studies investigating how to distribute the medical resource for patients with other diseases during the pandemic period.

Concept
The concept of our review will be the triage criteria, systems, or algorithms used for the allocation of medical resources to selected populations among patients with the pandemic disease.

Context
The context of our review will be the resource scarcity setting during disease pandemic like COVID-19.

Types of evidence resources
We will include studies reporting the tool, criteria, consensus, or recommendation to triage patients under the resource scarcity setting during the era of disease pandemic. We will also include studies investigating the efficiency of existing triage tools. We will consider any publication types as a potentially relevant study.

Information sources and search strategy
We will conduct the systematic search on MEDLINE, Embase and Cochrane controlled register of trials (CENTRAL). The detailed search strategy for MEDLINE is as below;
<u>MEDLINE via Pubmed</u>

#1 "Resource Allocation"[Mesh] OR Triage[Mesh] OR allocation*[tiab] OR triage*[tiab]
#2 pandemics[mesh] OR “disease outbreak”[tiab] OR pandemic*[tiab] OR outbreak*[tiab]
#3 #1 AND #2

Selection of sources of evidence

Two review authors will independently review titles and abstracts identified by the search strategy. The same two authors will retrieve the full text of potentially relevant studies and independently will assess the full text against the eligibility criteria outlined in Criteria for considering studies for this review. Differences will be resolved by consensus.

Data extraction

Two review authors will use a structured data extraction form to collect the data from the included studies independently. We will extract the following information: author, country, year, study design, setting, what population the criteria for (e.g., SARS, MARS, COVID-19 or not specified, other pandemic diseases), for what medical resource being got triage, elements of criteria, the timing of use, method of criteria development (e.g., using the Delphi method or not, or the number of the members developed the criteria), patient involvement for criteria development, and publication status. We will further extract the following information if the triage was implemented in a pandemic situation; the number of patients got to triage, selected or excluded, patient characteristics including age and sex, efficiency as the authors defined, and adverse events. We will resolve any disagreements through discussion or by consulting a third review author if necessary.

Analysis of the evidence

We will summarize the studies based on the characteristics and efficiency (if applicable) of triages as a format of diagram or tabular form according to the objective of this review.

References

1. COVID-19 Map. Johns Hopkins Coronavirus Resource Center. Available from: https://coronavirus.jhu.edu/map.html (Accessed 16 Apr 2020).

2. Remuzzi A, Remuzzi G. COVID-19 and Italy: what next? Lancet 2020: 395:1225-1228.

3. Emanuel EJ, Persad G, Upshur R, Thome B, Parker M, Glickman A, et al. Fair allocation of scarce medical resources in the time of Covid-19. N Engl J Med 2020 Mar 23 [Epub ahead of print]

4. Tricco AC, Lillie E, Zarin W, O’Brien KK, Colquhoun H, Levac D, et al. PRISMA Extension for Scoping Reviews (PRISMA-ScR): Checklist and Explanation. Ann Intern Med. 2018;169:467–73.

5. Peters MDJ, Godfrey C, McInerney P et al. Ch 11. Scoping reviews In: Aromataris E, Munn Z (Editors).Joanna Briggs Institute Reviewer's Manual. The Joanna Briggs Institute, 2017. Available from https://reviewersmanual.joannabriggs.org/